



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - *Prova scritta preliminare del 11/01/2018*

PROBLEM - We observe N samples of the following discrete-time real random process:

$$X[n] = A \cdot s[n] + W[n], \quad n = 0, 1, \dots, N-1,$$

where A is the amplitude of the Signal of Interest (SOI) and it is modelled as a random parameter, in particular it is modelled as a zero mean Gaussian random variable with variance σ_A^2 , $s[n]$ is the signal shape, perfectly known, its energy is given by $E_s = \sum_{n=0}^{N-1} s^2[n] = N$, $W[n]$ is additive white Gaussian noise (AWGN) with Power Spectral Density (PSD) $S_w(e^{j2\pi f}) = \sigma_w^2$, where σ_w^2 is a-priori known.

(1) Derive the joint probability density function (pdf) of the received data vector $\mathbf{X} \triangleq [X[0] \ X[1] \ \dots \ X[N-1]]^T$ and of the random parameter A (this pdf can be expressed either in scalar form or in vector form, provide both). Then, derive the marginal pdf of the received data vector $\mathbf{X}$.

(2) Under the assumption that σ_w^2 is a-priori known, derive the Maximum A-Posteriori (MAP) estimator of the amplitude A , the pdf of the estimation error $\varepsilon \triangleq A - \hat{A}_{MAP}$ and the Mean Square Error (MSE) of the estimator, $MSE \triangleq E\{\varepsilon^2\}$. Express these quantities in terms of the SNR $\gamma \triangleq \sigma_A^2 / \sigma_w^2$ and establish whether the estimator is biased and consistent.

(3) Derive the Bayesian Cramèr-Rao lower bound (CRB), compare it with the MSE of the MAP estimator and establish whether the estimator is efficient or not.

Hint: Woodbury's identity: $(\mathbf{I} + \gamma \mathbf{ss}^T)^{-1} = \mathbf{I} - \frac{\gamma}{1 + \gamma \mathbf{ss}^T} \mathbf{ss}^T$



Solution

(1) Let's define $X_n \triangleq X[n]$ and $s_n \triangleq s[n]$. The pdf of the observed data vector conditioned to A is given by:

$$X_n|A \in N(A s_n, \sigma_w^2), \quad \{X_n|A\}_{n=0}^{N-1} \text{ conditionally independent} \Rightarrow \mathbf{X}|A \in N(\mathbf{A}\mathbf{s}, \sigma_w^2 \mathbf{I})$$

$$A \in N(0, \sigma_A^2)$$

The pdf in scalar format is:

$$\begin{aligned} f_{X,A}(\mathbf{x}, a) &= f_{X|A}(\mathbf{x}|a) f_A(a) = \prod_{n=0}^{N-1} f_{X|A}(x_n|a) f_A(a) \\ &= \left(\frac{1}{\sqrt{2\pi\sigma_w^2}} \right)^N \exp\left(-\frac{1}{2\sigma_w^2} \sum_{n=0}^{N-1} (x_n - a \cdot s_n)^2 \right) \cdot \frac{1}{\sqrt{2\pi\sigma_A^2}} \exp\left(-\frac{a^2}{2\sigma_A^2} \right) \end{aligned}$$

In vector format:

$$\begin{aligned} f_{X,A}(\mathbf{x}, a) &= f_{X|A}(\mathbf{x}|a) f_A(a) \\ &= \frac{1}{(2\pi\sigma_w^2)^{N/2}} \exp\left(-\frac{1}{2\sigma_w^2} (\mathbf{x} - \mathbf{A}\mathbf{s})^T (\mathbf{x} - \mathbf{A}\mathbf{s}) \right) \cdot \frac{1}{\sqrt{2\pi\sigma_A^2}} \exp\left(-\frac{a^2}{2\sigma_A^2} \right) \end{aligned}$$

Pdf of $\mathbf{X}$ in vector format: $\mathbf{X} = \mathbf{A}\mathbf{s} + \mathbf{W}$, $\mathbf{X}$ is a Gaussian vector, since $\mathbf{A}$ and $\mathbf{W}$ are Gaussian distributed: $\mathbf{X} \in N(\boldsymbol{\eta}_X, \mathbf{C}_X)$

$$\boldsymbol{\eta}_X = E\{\mathbf{X}\} = E\{\mathbf{A}\mathbf{s} + \mathbf{W}\} = E\{\mathbf{A}\}\mathbf{s} + E\{\mathbf{W}\} = \mathbf{0}$$

$$\mathbf{C}_X = E\{\mathbf{X}\mathbf{X}^T\} = E\{A^2\}\mathbf{s}\mathbf{s}^T + E\{\mathbf{W}\mathbf{W}^T\} = \sigma_A^2 \mathbf{s}\mathbf{s}^T + \sigma_w^2 \mathbf{I}$$



$$f_X(\mathbf{x}) = \frac{1}{(2\pi)^{N/2} \sqrt{|\sigma_A^2 \mathbf{ss}^T + \sigma_w^2 \mathbf{I}|}} \exp\left(-\frac{1}{2} \mathbf{x}^T (\sigma_A^2 \mathbf{ss}^T + \sigma_w^2 \mathbf{I})^{-1} \mathbf{x}\right)$$

$$= \frac{1}{(2\pi\sigma_w^2)^{N/2} \sqrt{|\gamma \mathbf{ss}^T + \mathbf{I}|}} \exp\left(-\frac{1}{2\sigma_w^2} \mathbf{x}^T (\gamma \mathbf{ss}^T + \mathbf{I})^{-1} \mathbf{x}\right)$$

where $\gamma \triangleq \sigma_A^2 / \sigma_w^2$

The inverse of the covariance matrix can be derived by using the *Woodbury's identity*:

$$(\mathbf{I} + \gamma \mathbf{ss}^T)^{-1} = \mathbf{I} - \frac{\gamma}{1 + \gamma \mathbf{ss}^T} \mathbf{ss}^T = \mathbf{I} - \frac{\gamma}{1 + N\gamma} \mathbf{ss}^T$$

The determinant is given by: $|\gamma \mathbf{ss}^T + \mathbf{I}| = 1 + N\gamma$. Hence, the pdf of $\mathbf{X}$ is given by:

$$f_X(\mathbf{x}) = \frac{1}{(2\pi\sigma_w^2)^{N/2} \sqrt{1 + N\gamma}} \exp\left(-\frac{1}{2\sigma_w^2} \mathbf{x}^T \left(\mathbf{I} - \frac{\gamma}{1 + N\gamma} \mathbf{ss}^T\right) \mathbf{x}\right)$$

(2) MAP estimator of A :

$$\hat{A}_{MAP} = \arg \max_a (\ln f_{X,A}(\mathbf{x}, a)) = \arg \max_a (\ln f_{X|A}(\mathbf{x}|a) + \ln f_A(a))$$

$$\ln f_{X|A}(\mathbf{x}|a) + \ln f_A(a) = \kappa - \frac{1}{2\sigma_w^2} \sum_{n=0}^{N-1} (x_n - a \cdot s_n)^2 - \frac{a^2}{2\sigma_A^2}$$

$$\frac{d}{da} (\ln f_{X|A}(\mathbf{x}|a) + \ln f_A(a)) = \frac{1}{\sigma_w^2} \sum_{n=0}^{N-1} s_n (x_n - a \cdot s_n) - \frac{a}{\sigma_A^2} = \frac{1}{\sigma_w^2} \left[\sum_{n=0}^{N-1} s_n x_n - aN - \frac{1}{\gamma} a \right] = 0$$

$$\Rightarrow \hat{A}_{MAP} = \frac{\gamma}{1 + N\gamma} \sum_{n=0}^{N-1} s_n x_n = \frac{\alpha}{N} \sum_{n=0}^{N-1} s_n x_n = \frac{\alpha}{N} \mathbf{s}^T \mathbf{x}, \quad \text{where } \alpha \triangleq \frac{N\gamma}{1 + N\gamma}$$

Since the estimate is a linear function of the Gaussian distributed data vector $\mathbf{X}$, the estimate is Gaussian distributed, and consequently the estimation error is **Gaussian distributed**.



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The mean and variance of the error are given by:

$$E\{\hat{A}_{MAP}\} = E\left\{\frac{\alpha}{N} \mathbf{s}^T \mathbf{x}\right\} = \frac{\alpha}{N} \mathbf{s}^T E\{\mathbf{x}\} = \frac{\alpha}{N} \mathbf{s}^T \boldsymbol{\eta}_x = 0 \Rightarrow E\{A - \hat{A}_{MAP}\} = E\{A\} - E\{\hat{A}_{MAP}\} = 0$$

Hence, the estimator is **unbiased**.

$$\begin{aligned} MSE\{\hat{A}_{MAP}\} &= E\left\{\left(A - \hat{A}_{MAP}\right)^2\right\} = E\left\{\left(A - \frac{\alpha}{N} \mathbf{s}^T \mathbf{x}\right)^2\right\} = E\left\{\left(A - \frac{\alpha A}{N} \mathbf{s}^T \mathbf{s} + \frac{\alpha}{N} \mathbf{s}^T \mathbf{W}\right)^2\right\} \\ &= E\left\{\left((1-\alpha)A + \frac{\alpha}{N} \mathbf{s}^T \mathbf{W}\right)^2\right\} = (1-\alpha)^2 \sigma_A^2 + \frac{\alpha^2}{N^2} \mathbf{s}^T E\{\mathbf{W}\mathbf{W}^T\} \mathbf{s} = (1-\alpha)^2 \sigma_A^2 + \frac{\alpha^2}{N} \sigma_W^2 \\ &= \frac{1}{(1+N\gamma)^2} \sigma_A^2 + \frac{N\gamma^2}{(1+N\gamma)^2} \sigma_W^2 = \frac{\sigma_W^2 \gamma}{1+N\gamma} = \alpha \frac{\sigma_W^2}{N} \end{aligned}$$

$$\lim_{N \rightarrow \infty} MSE\{\hat{A}_{MAP}\} = \lim_{N \rightarrow \infty} \alpha \cdot \lim_{N \rightarrow \infty} \frac{\sigma_W^2}{N} = 1 \cdot 0 = 0$$

Hence, the estimator is **consistent**.

$$\varepsilon = A - \hat{A}_{MAP} \in N\left(0, \alpha \frac{\sigma_W^2}{N}\right)$$

(3) BCRB on the estimate of A:

$$\frac{d^2 \ln f_{X,A}(\mathbf{x}, a)}{da^2} = \frac{d^2}{da^2} \left(\ln f_{X|A}(\mathbf{x}|a) + \ln f_A(a) \right) = \frac{1}{\sigma_W^2} \cdot \frac{d}{da} \left[\sum_{n=0}^{N-1} s_n x_n - aN - \frac{1}{\gamma} a \right] = -\frac{1+N\gamma}{\gamma \sigma_W^2}$$

$$BCRB(A) = \frac{1}{-E\left\{\frac{d^2 \ln f_{X,A}(\mathbf{x}, a)}{da^2}\right\}} = \frac{\gamma \sigma_W^2}{1+N\gamma} = \alpha \frac{\sigma_W^2}{N} = MSE\{\hat{A}_{MAP}\}$$

Hence, the estimator is **efficient**.